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Preface

The **NeSS International Conference (2026)** brings together researchers, academicians, practitioners, and students from around the world to share recent advances, emerging trends, innovative methodologies, and practical applications on Statistics and Data Sciences. The conference provides a platform for exchanging ideas, promoting interdisciplinary collaboration, and fostering professional networks among participants.

This abstract book contains the accepted abstracts presented during the conference, including keynote seasons, invited talks, oral presentations, and poster presentations. The abstracts reflect the diversity of current research and demonstrate the growing impact of statistics, data science, artificial intelligence, machine learning, and quantitative methodologies across a wide range of disciplines.

The organizing committee sincerely appreciates the valuable contributions of all authors, reviewers, session chairs, keynote speakers, invited speakers, sponsors, and volunteers whose dedication and support made this conference possible. Their commitment has ensured the high academic quality of the conference programme.

We hope this abstract book will serve as a useful reference for participants during the conference and continue to be a valuable resource for future research collaborations.

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Prof. Dr. Bijay Lal Pradhan
Prof. Dr. Rabindra Kayastha

NeSS International Conference (2026)
Kathmandu, Nepal
20-21 September 2026

Message from the Conference Chair

It is my great pleasure to welcome all participants to **NeSS International Conference (2026)** , held in **Kathmandu, Nepal** from **20-21 September 2026**.

This conference provides an excellent platform for researchers, academicians, professionals, and students to present their latest research findings, exchange innovative ideas, and establish meaningful collaborations. As statistical science continues to evolve rapidly in response to emerging global challenges, meetings such as this play an important role in promoting scientific discussion and interdisciplinary research.

The scientific programme has been carefully designed to include keynote lectures, invited talks, oral presentations, and poster sessions covering a broad spectrum of topics in statistics, data science, artificial intelligence, machine learning, biostatistics, official statistics, actuarial science, and related fields.

This abstract book showcases the accepted contributions of the conference and reflects the dedication and scholarly efforts of the authors. I sincerely thank all contributors, reviewers, session chairs, members of the organizing committee, sponsors, and volunteers for their valuable support and commitment in making this conference a success.

I wish all participants a productive conference, engaging discussions, and an enjoyable stay in Kathmandu, Nepal. May this event inspire new ideas, strengthen existing collaborations, and create opportunities for future research partnerships.

Prof. Dr. Ram Prasad Khatiwada
Conference Chair
NeSS International Conference (2026)

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K001

Phase-wise Analysis of the Bowling Performance of Cricketers in the Nepal Premier League

Prof. Dibyojyoti Bhattacharjee[1],

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Keywords: Cricket Analytics, Data Mining in Sports, Combined Bowling Rate, Format Specific Bowling Performance Measure, Franchisee Cricket, Nepal Premier League

Cricket being a data rich sports and data management becoming simpler owing to the advances in computer technology, the sports have now become a data-driven discipline. Simultaneously, cricket has also seen mushrooming of different franchise league throughout the globe which has enhanced the need of precise player analytics. This presentation on the basis of bowling data collected from the first two seasons of the Nepal Premier League (a franchise-based Twenty20 cricket tournament established in 2024) quantifies the performance of bowlers in three distinct match phases viz. Powerplay, middle and death overs. Two different measures of bowling performance- the Combined Bowling Rate (CBR) and the Format Specific Bowling Performance Measure (FSB). The distributional pattern of the two measures were later utilized to attain practical benchmark for superlative bowlers in the different phases of a Twenty20 innings. The exercise provides the coaches with a strategic framework for the rotation of bowlers during live match and helps the franchises in making informed and cost effective bidding during player auction so that the squad has specialist bowlers for all different match situations.

K002

Title

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Keywords: ...

...

K003

Title

Prof. Dr. Sudan Jha[1],

Member, National Planning commission

Keywords: ...

2 Oral Presentation Abstracts

T001

From Anti-Spoofing to Trustworthy Audio Forensics: A Bibliometric and Science-Mapping Review of Audio Deepfake Detection

Mr. Prawesh Bhandari[1], Ms. Prashansha Neupane[1], Ms. Arju Marasini[2], Mr. Bishwas Sharma[3]

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Keywords: Anti-spoofing; Audio deepfake detection; Bibliometric analysis; Science mapping; Synthetic speech.

This study maps the intellectual structure, knowledge base, and emerging frontiers of audio-deepfake detection, addressing the absence of an integrated account of the field's evolution, influence, collaboration, and thematic change. A PRISMA-informed bibliometric and science-mapping review analysed 945 deduplicated Scopus and Web of Science records (2020-2026; one early-online 2027 record). Performance analysis, citation and co-citation analysis, bibliographic coupling, collaboration mapping, keyword co-occurrence, thematic mapping, and trend analysis were conducted using bibliometrix/Biblioshiny. The field expanded rapidly (56.9% complete-year annual growth) but remains conference-dominated and only selectively internationalised. ASVspoof and a small set of benchmark-centred studies form the intellectual core, while recent research is shifting towards self-supervised learning, cross-dataset generalisation, real-time detection, source tracing, and explainable forensics. Persistent gaps concern multilingual and low-resource data, robustness to unseen generators and channels, privacy, fairness, and operational validation. The field is moving from benchmark optimisation towards auditable, provenance-aware, and deployment-ready detection. Progress depends on standardised cross-language evaluation, representative real-world datasets, transparent uncertainty reporting, and closer integration of technical, ethical, and governance perspectives.

T002

Unequal Education-to-Work Pathways: Gendered Labour-Force Exclusion and SDG Interlinkages in Nepal

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Keywords: Labour-force exclusion; Education-to-work conversion; Gender inequality; Sustainable Development Goals

This study examines whether education reduces labour-force exclusion in Nepal, whether its effects differ by gender, and whether excluded adults include a discouraged but potentially employable group. Using 27,011 adults from the nationally representative Nepal Living Standards Survey 2022/23, the study applies survey-weighted descriptive analysis, logistic regression, education-by-gender interaction models, predicted probabilities, and a sub-analysis of discouraged-like exclusion. Labour-force exclusion affected 65.0% of adults. Education showed a clear threshold effect: tertiary education offered the strongest protection, while lower and intermediate levels produced limited gains. Women remained more likely than men to be excluded at every education level; even tertiary-educated women had higher predicted exclusion than men with no schooling. Among excluded adults, higher education was linked to greater willingness to work if suitable jobs were available, indicating hidden labour potential. Education alone does not guarantee access to the labor market. Nepal needs to have demand-driven vocational education system, employment opportunities, childcare provision, safe means of transport, placement service provision, and gender-aware labor market policies to conform to SDG goals 4, 5, 8, and 10. The present study merges the idea of gendered education, work transition, labor force exclusion, and discouraged labor force participation into one nationally representative approach.

T003

An Enhanced Air Quality Index for Multi-Pollutant Air Quality Assessment

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Keywords: Air Pollution, Composite Index, Environmental Monitoring, Classification, Sensitivity Analysis

Air Quality Index (AQI) is an essential tool for communicating ambient air quality and its associated health risks. However, conventional AQI frameworks often rely on the maximum sub-index approach, which may not adequately capture the combined influence of multiple pollutants. This study proposes an improved AQI by integrating normalized pollutant concentrations using eight weighted aggregation techniques: Weighted Arithmetic Mean, Weighted Geometric Mean, Weighted Harmonic Mean, Principal Component Analysis (PCA) Index, Entropy Index, Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Ordered Weighted Averaging (OWA), and Fuzzy Aggregation. To evaluate the robustness of these methods, a One-at-a-Time (OAT) sensitivity analysis was conducted by perturbing individual pollutant concentrations while keeping the remaining pollutants unchanged. The results revealed that all aggregation methods exhibited relatively low sensitivity to input perturbations, indicating satisfactory stability. Among them, the Weighted Harmonic Mean demonstrated the lowest average sensitivity index (0.5040%), signifying the highest robustness and reliability, whereas TOPSIS showed the greatest sensitivity to changes in pollutant values. A novel six-category AQI classification scheme, based on the the Weighted Harmonic Mean is subsequently established while preserving the familiar structure of the existing AQI models This framework offers a reliable alternative to conventional AQI methods, least effected by the extreme observations, and can support improved environmental monitoring, public health assessment, and air quality management.

T004

Distributional and Stochastic Properties of Concomitants of Order Statistics from the Clayton Copula-Based Bivariate Gamma Distribution

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Keywords: Moment generating function; survival function; stochastic ordering; incomplete gamma function.

This paper investigates the distributional properties of concomitants of order statistics arising from a bivariate Gamma distribution constructed via the Clayton copula, referred to as the Clayton Copula-based Bivariate Gamma (CBG) distribution. Copula-based approaches offer a flexible and powerful framework for modeling dependence structures between marginal distributions. In this study, the joint and marginal probability density functions of the bivariate model, the conditional distributions, and the distribution of the r^{th} order statistic is derived. Closed-form and integral expressions for the density of the r^{th} concomitant are established, along with a finite-sum closed-form representation when the shape parameter is an integer. The moment generating function (MGF) and the first (mean) and higher order moments of the concomitant of order statistics are derived analytically. A stochastic ordering result is established, showing that higher-order concomitants are stochastically larger. The survival function of the concomitant is also obtained and analysed under varying degree of dependence. Graphical illustrations confirm the theoretical findings and highlight the versatile tail behaviour induced by the Clayton copula dependence structure.

T005

Consumer Shopping Culture in Marts and Departmental Stores: Evidence from Ward No. 9, Kathmandu Metropolitan City

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Keywords: Shopping Culture, Consumer Behaviour, Departmental Store, Retail Marketing, Customer Satisfaction, Kathmandu Valley

The quick expansion of supermarkets and departmental stores has ended up changing the shopping culture for city consumers in Nepal, sort of in a noticeable way. Before, most people relied on small neighbourhood grocery stores, but now those kinds of practices are getting replaced more and more by organized retail outlets. These newer places provide a bigger range of goods, competitive rates, frequent promotional schemes, and what many customers feel is a better overall shopping experience.

This study is meant to look into how customers actually shop when they visit marts and departmental stores in Ward No. 9, Kathmandu Metropolitan City. More precisely, it explores how the store environment, product quality, pricing strategies, promotional activities, customer service, convenience, and technological facilities affect consumers' shopping behaviour, and also their eventual purchase decisions. This study uses a quantitative approach based on the positivist paradigm and a descriptive plus analytical research design. The primary data will be gathered from customers via a structured questionnaire, with a five-point Likert scale that keeps things consistent. Participants will be chosen using systematic random sampling, and that should subtly reduce bias. To make sense of the responses, descriptive statistics will be applied, along with reliability analysis (Cronbach's Alpha), exploratory factor analysis, correlation testing and then multiple regression analysis will also be used.

Overall, the results are anticipated to show the main drivers that shape shopping culture inside contemporary retail outlets, and then offer workable suggestions for retailers so they can improve customer satisfaction, build loyalty, and stay competitive. In addition, the research is expected to add value to the continuing body of work on consumer behaviour and organised retailing in Nepal, especially within the local context.

T006

Comparison of the predictive performance of machine learning models for predicting low birthweight among live births in Nepal using data from the Nepal Demographic and Health Survey 2022

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Keywords: Low birth weight, Machine learning, SMOTE, NDHS

Low birth weight (LBW) is a major public health problem in Nepal and is associated with increased neonatal mortality and long-term adverse health outcomes. Machine learning (ML) offers a promising approach for early prediction of LBW, but evidence from Nepal is limited. This study compared the predictive performance of different ML models using nationally representative data. This study used data from the 2022 Nepal Demographic and Health Survey. After applying the exclusion criteria and handling missing data, 1,390 live births were included. Twenty-one(21) predictor variables were included based on literature review. Five machine learning models: logistic regression (LR), decision tree (DT), random forest (RF), support vector machine (SVM), and extreme gradient boosting (XGBoost) were developed and compared. The dataset was randomly split into training (80%) and testing (20%) sets. To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training set. Model performance was evaluated on the testing set using accuracy, sensitivity, specificity, precision, F1-score, and the area under the receiver operating characteristic curve (AUC). Predictive performance varied across the machine learning models. There was variation in the discriminative performance between the models before and after addressing class imbalance using SMOTE with RF demonstrating highest performance and SVM the greatest sensitivity. Variable importance analysis was also performed which identified variables with birth order being the most influential predictor in this analysis alongwith other factors such as geography and socioeconomic status related variables. The predictive performance of the various models varied and the effect of using SMOTE was also variable among the models. SMOTE based machine learning may support accurate detection of pregnancies with high risk of LBW.

T007

Classical and Objective Bayesian Approaches to Estimating the Process Capability Index Cpc for the Dhillon Process Distribution

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Keywords: Process capability index; Bootstrap confidence intervals; Product spacing function, Bayesian estimation methods; Monte Carlo simulation.

The process capability index (PCI) is a valuable tool for evaluating how effectively a product meets customer expectations and for assessing overall process performance. When the quality characteristics of a process follow a normal distribution, conventional PCIs typically provide reliable and accurate results. However,these traditional indices may lead to incorrect conclusions when applied to non-normally distributed processes, potentially resulting in flawed decisions. In light of this challenge, the present study focuses on the process capability index Cpc, which remains suitable for both normal and non-normal process distributions. Specifically, we consider a process that follows the Dhillon distribution. Further, classical estimation alongwith classical confidence intervals and bootstrap confidence intervals for Cpc are constructed and evaluated based on their average widths and coverage probabilities. Additionally, Bayesian estimation of Cpc is carried out using both the product spacing and likelihood functions under symmetric and asymmetric loss functions assuming objective priors for the model parameters through Markov Chain Monte Carlo algorithm. A detailed Monte Carlo simulation study is conducted to evaluate the performance of the proposed estimators. Finally, the methodology is illustrated using three real datasets, thereby confirming the practicality, flexibility, and reliability of the proposed inferential framework for process capability evaluation.

T008

Climate Stressors and Dairy Milk Production: Eastern Nepalese Farmers' Perception

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Keywords: Farmers' perception, adaptation, climate stress, disease prevalence, Koshi Province, mixed methods

Climate change has threatened livestock farming, influencing food security, rural livelihoods, and the national economy. This study assessed farmers' perception on 16 factors associated with four each from four domains: milk production, reproduction, health management, and methane production affecting dairy milk production (average milk production in litres per cattle per year). A total of 223 dairy farmers from Jhapa, Morang, and Sunsari districts in Koshi Province were surveyed in May 2024 using a structured questionnaire with a five-point Likert scale (5=very highly effect to 1= very low effect). Sex of the household head (female as a dummy variable with 1 and 0 otherwise) was also considered as a determining factor. The ordinary least square (OLS) regression models showed that the thermal stress in reproduction had the strongest positive effect on milk production, with each unit increase associated with a 0.455 increase in milk yield. Disease prevalence had the positive effect on milk production, with each unit decrease associated with a 0.319 increase in milk production. The supply of leguminous forage and concentrate on methane reduction had the positive effect on milk production, with each unit increase associated with a 0.319 increase in milk production. Results showed significant negative effect of the prevalence of vectors borne to animal health management system on milk production, with each unit increase associated with a 0.286 decline in milk production. The study presents a framework linking farmer perceptions with measured impacts and highlights the need for targeted interventions. Strengthening adaptive capacity requires improved disease prevention through vaccination, better housing to reduce heat stress, tick and vector control programs, and the development of climate-resilient leguminous forage varieties. These findings should guide extension workers, academic institutions, program planners, and policymakers in designing climate-resilient livestock strategies.

T009

A Machine Learning-Based Framework for Tomato Maturity and Sweetness Classification in Smart Agriculture

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Keywords: Tomato quality assessment, Maturity classification, Sweetness classification, ResNet50, Principal Component Analysis, Machine learning.

Smart agriculture has emerged as a promising approach for improving crop quality assessment through the integration of artificial intelligence, computer vision, and machine learning. Horticultural crops play a vital role in human nutrition, making accurate quality assessment essential for enhancing consumer satisfaction, market value, and post-harvest management. Among these crops, tomato is one of the most widely consumed vegetables worldwide, and its quality is primarily determined by maturity stage and sweetness. Conventional methods for evaluating these quality attributes are destructive, labor-intensive, and time-consuming, limiting their suitability for rapid and non-destructive assessment. This study proposes a machine learning-based framework for simultaneous tomato maturity and sweetness classification using feature fusion. A data-set comprising 450 tomatoes and approximately 1,400 images representing three different stages of maturity, namely Light Red, Turning Red, and Ripe Red, was acquired using a smartphone camera under controlled imaging conditions. Handcrafted colour and texture features were combined with deep features extracted from a pre-trained ResNet50 model after dimensionality reduction using Principal Component Analysis (PCA) to generate a fused feature representation. Four machine learning algorithms, namely Random Forest, XG-Boost, Support Vector Machine, and Multi-layer Perceptron (MLP), were evaluated for maturity classification, where the MLP classifier achieved the highest accuracy of 91.93%. To assess internal quality, measured Brix values were grouped into Low and High sweetness categories using the k-Means clustering algorithm. The predicted maturity stage was incorporated as an additional feature for sweetness classification, resulting in a 164-dimensional feature vector, with the Random Forest classifier achieving the best accuracy of 62.46%. Furthermore, a prototype real-time tomato grading system was developed to demonstrate the practical applicability of the proposed framework. The findings demonstrate that integrating feature fusion and machine learning provides an effective non-destructive approach for tomato quality assessment and supports the development of intelligent decision-support systems for smart agriculture and post-harvest quality management.

T010

Truncated Shanker Distribution with Properties and Applications

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Keywords: Truncation; Lifetime Distribution; Maximum Likelihood Estimation; Statistical properties; goodness of fit

Lifetime distributions play a fundamental role in reliability engineering, survival analysis, actuarial science, and biomedical research for modelling positive continuous data. However, in many practical situations, observations are available only within a restricted range due to design constraints, detection limits, or data collection procedures, making truncated models more appropriate than their parent distributions. Motivated by this need, the present study introduces the Truncated Shanker Distribution (TShD) by truncating the Shanker distribution over a specified interval. The proposed model includes left-truncated, right-truncated, and doubly truncated forms, thereby providing greater flexibility for modelling bounded lifetime data. Several statistical properties of the proposed distribution are investigated, including the probability density function, cumulative distribution function, survival function, hazard rate function, reverse hazard rate function, mean residual life function, and the first four raw moments. The model parameters are estimated using the Maximum Likelihood Estimation (MLE) method, and the likelihood equations are derived explicitly.

T011

Quantifying the Quality of Water of Some Naturally Occurring Perennial Lakes of South Assam, India

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Keywords: Composite Index, Water Quality Index, Seasonal Variation, Sensitivity Analysis.

Wetlands are transitional zones between terrestrial and aquatic systems that remain saturated due to high groundwater or surface water for part or all of the year. Water quality is an important environmental subject and dynamic factor in wetland ecosystem productivity, which can be determined by analysing the water physically, chemically and biologically. South Assam of India, which is a valley in the foothills of the Barail Range of hills, receives enormous rainfall both pre- and during the monsoon, leading to several naturally occurring perennial lakes in the region. These lakes are the source of water for domestic use by the residents, including drinking, washing, cleaning, bathing, etc., in addition to the use of this water for irrigation. But, proper testing of the water in terms of its physical, chemical and biological properties and their changes across the different seasons of the year is rarely attempted. In this paper, the researchers attempted to quantify the water quality and their variability across the different months of the year using several composite indices for three perennial lakes of South Assam, India. The indices are then compared using sensitivity and compatibility models.

T012

Inference on stress-strength reliability using product spacing function for the inverted exponentiated gamma distribution with application to the electrode breakdown times data

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Keywords: Inverted exponentiated gamma distribution, Stress-Strength reliability, Maximum product spacing estimation, Bayesian inference; Bayes computation

In this paper, point and interval estimation procedures for the stress–strength reliability (SSR) parameter are developed based on the inverted exponentiated gamma distribution (IEGD). The IEGD is proposed as a new and flexible lifetime distribution that exhibits desirable characteristics for modeling reliability and survival data. Classical estimation procedures for the SSR parameter are formulated using both the maximum likelihood estimation (MLE) and maximum product spacing estimation (MPSE) methods. For interval estimation, asymptotic confidence intervals (ACIs) are derived based on MLE and MPSE, along with three bootstrap confidence intervals namely standard (s-boot), percentile (p-boot), and t-bootstrap (t-boot). Furthermore, a comprehensive Bayesian framework is developed by incorporating independent gamma flexible priors under a squared error loss function, based on both the usual likelihood function and the product spacing function. Bayesian credible intervals for the SSR parameter are constructed and compared with their classical counterparts. The comparative analysis reveals that both classical and Bayesian estimators, as well as their associated intervals, exhibit satisfactory efficiency and coverage characteristics. Finally, the practical utility of the proposed methods is illustrated through a real data application, demonstrating the suitability and effectiveness of the IEGD for reliability analysis.

T013

Do More Words Change Minds? Tracing Sentiment Transitions in Nepali Political Tweets

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Keywords: Sentiment Analysis, Political Tweets, Low resource Language

Political tweets often express opinions about political figures, parties, and events. However, it remains unclear how people form sentiment judgments when only part of a message is initially visible. This study examines how these judgments change as more words are revealed in Nepali political tweets. We use sentiment-labeled tweets from the Sentiment of Election-Based Nepali Tweets dataset, collected around Nepal's 2022 election. Initially, 50% of the words in each tweet are randomly masked. Participants rate the visible text on a continuous sentiment scale from 0 to 5, where 0 is the most negative and 5 is the most positive. Participants then gradually reveal the masked words and may update their rating after each reveal. We record the sequence of ratings, number of reveals, time between reveals, and basic demographic information. The final analytic dataset includes 7,028 submissions from 169 participants across 1,523 unique tweets. The results show that initial sentiment judgments strongly shape later ratings. Although the average final sentiment increased by 0.32 points, the median change was 0.00, indicating substantial persistence of first impressions. Overall, 44.3% of participant-tweet pairs became more positive, 22.7% became more negative, and 33.0% remained unchanged. Compared with the original sentiment labels, the participant's label agreement was 41.7% for the initial masked text, increasing to 52.1% after the final word was revealed; however, this did not fully remove the ambiguity. Additionally, the sentiment state transitions also became more stable after word revelation, with the same category transition probability increasing from 64.9% before any reveal to 78.0% after at least one reveal. Word-level analysis showed that masked and revealed terms were concentrated around election-related concepts such as , , , , and . These findings demonstrate that progressive word reveal is a useful method for studying how partial information, early impressions, and newly revealed linguistic cues interact in human sentiment interpretation, especially for political text in low-resource languages such as Nepali.

T014

Knowledge, Attitude and Preventive Practices Regarding Dengue Fever Among Residents of Slum Area in Butwal Sub-Metropolitan City, Rupandehi

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Keywords: dengue fever; knowledge, attitude and practice; urban slum; Nepal; ordinal logistic regression; knowledge–practice gap

Dengue fever is one of the fastest-spreading mosquito-borne viral diseases in Nepal, with urban slum communities facing increased risk because of overcrowding, unsafe water storage, and inadequate waste management. This community-based cross-sectional study assessed dengue-related knowledge, attitudes, and practices (KAP) among 401 adult residents of a slum area in Butwal Sub-Metropolitan City, Rupandehi District, Nepal, using a structured questionnaire adapted from the WHO KAP Survey Resource Pack. Data were analyzed in R using Bloom's percentage cut-offs, chi-square and Spearman's correlation tests, and logistic regression models.

Almost all participants (99.8%) had heard of dengue; however, knowledge was predominantly moderate (47.6%), with only 32.4% demonstrating good knowledge. Preventive practice was poor among 49.1% of respondents, while only 5.7% exhibited good practice. Knowledge was negatively correlated with preventive practice (Spearman's = 0.308, p ¡ 0.001), indicating that awareness did not necessarily translate into protective behaviour. Higher educational attainment was the strongest predictor of better knowledge, whereas lower educational attainment was unexpectedly associated with better preventive practices. Age, waste-disposal practices, and dengue history were significant predictors of attitude.

The findings highlight a substantial knowledge–practice gap among slum residents, suggesting that dengue control efforts should combine targeted behaviour-change interventions with improvements in water storage, sanitation, and waste-management infrastructure to promote effective disease prevention.

T015

Multivariate Statistical Modeling for the Analysis of NEPSE Stock Index

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Keywords: NEPSE, remittance, VECM, LSTM, macroeconomic variables, forecasting.

Nepal Stock Exchange (NEPSE) plays a pivotal role in Nepal's financial ecosystem nurturing economic growth, corporate development and investment opportunity. The study of macroeconomic variables impact on the NEPSE Index is significant as it provides keen insight for investors and policy makers to make informed decisions, enhancing market efficiency and economic stability. Acurate stock price prediction remains challenging because financial market exhibits noisy, volitile, complex multifactorial nature of local and global market. Still, building a strong model is now not so far, as driven by progress in AI, access to massive data, and increased computing capabilities. This study aims to construct a predictive model for NEPSE using several new headlines to predict where the closing price would rise or descend after each one-month moment direction. Utilizing a high-density, balanced monthly longitudinal dataset spanning a 15-year horizon (2010–2025, T = 192), the research architecture incorporates eight critical exploratory indicators: real Gross Domestic Product (GDP), broad money supply (M2), workers' remittances, domestic consumer price inflation, central bank discount rates, the US dollar exchange rate, international gold prices, and global crude oil shocks. The Johansen trace test confirms the existence of exactly one cointegrating (long-run equilibrium) relationship between NEPSE and the eight macroeconomic variables. The estimated long-run equilibrium relationship indicates that Broad Money Supply (BMS), Gold Price, Exchange Rate, Remittance have statistically significant long-run effects on NEPSE (p ¡ 0.05), whereas the interest rate, Inflation rate, GDP, Oil price does not have a statistically significant long-run effect (p ¿ 0.05). The short-run dynamics indicate that only the lagged change in remittance significantly influences the current month's change in NEPSE (p = 0.031); the short-run effects of the remaining variables are statistically insignificant. The VECM-selected variables achieved similar forecasting performance with a simpler and more stable model, supporting a parsimonious forecasting framework. Hyperparameter tuning improved the performance of all LSTM models, yielding comparable predictive accuracy ($R^2 \approx 0.75$–0.80) across different variable sets.

T016

Proportional Distribution of Childhood Cancer Types and Subtypes in Nepal: A Multicenter Hospital-Based Epidemiological Study

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Keywords: Neoplasms; Pediatrics; Epidemiology; Neoplasms by Histologic Type; Nepal

Childhood cancer is an important public health concern, yet evidence on its epidemiological distribution in Nepal remains limited. This study aimed to determine the proportional distribution of childhood cancer types and subtypes among pediatric patients in Nepal. A multicenter, hospital-based epidemiological study was conducted among children aged 18 years diagnosed with cancer at four major hospitals in Nepal during 2015-2025. Demographic and clinical data were extracted from medical records. Ethical approval was obtained from the Nepal Health Research Council (NHRC; Ref. No. 465), with permission from all participating hospitals. Descriptive statistics were used, with categorical variables summarized as frequencies and percentages and continuous variables as means and standard deviations. Among 2,180 pediatric cancer patients, leukemia was the most common cancer type (34.5%), followed by lymphoma (18.2%) and central nervous system tumors (12.7%). Cancer cases were more frequently observed among males than females, with higher proportions among children from marginalized caste communities and those residing in the Terai region. Cancer types and subtypes varied across demographic and geographic characteristics. Conclusion: Childhood cancer in Nepal shows a heterogeneous epidemiological distribution. These findings provide baseline evidence to strengthen cancer surveillance, support early diagnosis, guide pediatric oncology service planning, and promote equitable allocation of cancer care resources.

T017

Foreign Direct Investment and Economic Growth in Nepal: Evidencefrom ARDL and ARIMA Forecasting Models

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Keywords: Foreign Direct Investment; Economic Growth; ARDL; ARIMA; Forecasting, MAE,RMSE

This study investigates the relationship between foreign direct investment (FDI) and economic growth in Nepal using annual time-series data from 1960 to 2024. Employing the Autoregressive Distributed Lag (ARDL) bounds testing approach, the analysis examines both short-run dynamics and long-run equilibrium relationships between GDP growth and FDI. The empirical results reveal that FDI does not exert a statistically significant effect on economic growth in the short run. However, the bounds test confirms the existence of a stable long-run cointegrating relationship between FDI and GDP growth. The estimated error correction model indicates rapid adjustment toward long-run equilibrium, suggesting that deviations caused by short-term shocks are corrected swiftly, albeit with evidence of overshooting behavior. Structural break analysis further identifies regime changes in Nepal's growth trajectory. To assess predictive performance, the forecasting accuracy of the ARDL model is compared with that of an ARIMA framework using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). The results indicate that the ARDL model provides marginally superior forecasting accuracy relative to the ARIMA model. Overall, the findings suggest that while FDI does not stimulate immediate economic growth in Nepal, its long-run contribution depends critically on domestic absorptive capacity, institutional quality, and policy stability.

T018

Predicting the Real GDP and Its Growth Rate at Producer's Price for Next Five Years: Macroeconomic and Financial Situation

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Keywords: Real GDP, Economic Forecasting, ARIMA Model, Time Series Analysis, Monte Carlo Simulation

This study aims to forecast Nepal's Real Gross Domestic Product (GDP) growth rate at producer's price for the next five fiscal years (2025/26 to 2029/30) using the Box-Jenkins ARIMA methodology and Monte Carlo simulation techniques. Annual Real GDP data from fiscal year 1974/75 to 2024/25 obtained from Nepal Rastra Bank were analyzed. The study also examined the relationship between nominal and real GDP growth rates. Descriptive statistics, correlation analysis, linear regression, and time series forecasting methods were employed using R software. The findings reveal a moderate positive relationship between nominal and real GDP growth rates, with a Pearson correlation coefficient of 0.54. Regression results indicate that nominal GDP growth significantly predicts real GDP growth, although it explains only about 28.4% of its variation, suggesting a substantial influence of inflation and other economic factors. For forecasting, the ARIMA(2,0,0) model was identified as the most suitable model based on diagnostic tests and information criteria. The model projects a stable real GDP growth rate of approximately 4.3% to 4.4% annually over the next five years. To quantify forecast uncertainty, a Monte Carlo simulation with 2,000 future paths was conducted. The simulation estimated a 75.2% probability that real GDP growth will exceed 6% at least once during the forecast period and a 99.3% probability that real GDP will surpass 3,000,000 million rupees. The study concludes that Nepal's economy is expected to maintain moderate and stable growth in the coming years, although considerable uncertainty remains. The results highlight the importance of effective inflation management and macroeconomic stability to support sustainable real economic growth and improve the reliability of economic forecasting.

T019

Impact of Electronic Human Resource Management Practices on Employees' Job Satisfaction in the Banking Sector

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Keywords: Commercial banks, E-HRM practices, Employees' job satisfaction, Internet and productivity

The study aims to examine the impact of electronic human resource management (e-HRM) practices, focusing on e-recruitment, e-selection, e-compensation, e-learning and training, e-performance appraisal, and e-communication, on employees' satisfaction in commercial banks in Kathmandu. Structured questionnaires were used to collect data from 401 respondents working in commercial banks in Kathmandu District. The data were analyzed using SPSS version 25. Multiple regression analysis was conducted to predict employees' satisfaction from six independent variables: e-recruitment, e-selection, e-compensation, e-learning and training, e-performance appraisal, and e-communication. The findings indicate that employee satisfaction and e-HRM practices are positively correlated. The implementation of e-HRM systems in the banking industry can encourage employee achievement, which in turn leads to increased job satisfaction. E-HRM facilitates the achievement of strategic corporate goals by enhancing employee satisfaction. E-HRM practices also create a conducive work environment in which employees develop positive attitudes toward the organization. High job satisfaction among bank employees enhances productivity, organizational citizenship behavior, and organizational commitment, while decreasing absenteeism, employee turnover, and job-related stress. Hence, a high-performing e-HRM system may be useful for commercial banks to maintain harmony between employees' personal goals and organizational goals.

T020

Cubic Transmuted Nadarajah-Haghighi Distribution: Properties, Estimation and Application

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Keywords: Cubic transmutation map, Cubic rank transmuted Nadarajah-Haghighi distribution, Maximum product spacing, Reliability analysis, Markov chain Monte Carlo

This article suggest a new distribution, namely the cubic transmuted Nadarajah-Haghighi (CRT-NH) distribution by using cubic rank transmutation map. Various statistical properties such as pdf, reliability function, hazard function and its graphic, moment, moment generating function and order statistics of new distribution are examined. Estimation methods include maximum likelihood estimation (MLE), maximum product spacing (MPS), least squares estimation (LSE) for estimating the unknown shape, scale and transmutation parameters of the proposed model. A simulation study based on the Metropolis- Hastings Markov chain Monte Carlo (MCMC) algorithm is conducted to assess the performance of the estimators. Finally, a real data application is presented to demonstrate the flexibility and applicability of the proposed distribution.

T021

Adaptation Strategies as Mediators of the Effects of Economic, Socio-cultural, and Environmental Drivers on Migration Intentions in Nepal

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Keywords: Migration intention, Economic, Socio-cultural, Environmental impact, Adaptation

Migration has become one of the most pressing global issues of the twenty-first century. People's migration intentions are shaped by economic, sociocultural, demographic, and environmental factors. This study investigates how perceptions of the economic, sociocultural, and environmental impacts of migration influence individuals; intentions to migrate as an adaptive response to climate change in Gandaki Province, Nepal. A survey was conducted among 493 residents in four districts (Baglung, Myagdi, Mustang, and Lamjung) of Gandaki Province to explore residents; perceptions of the drivers of migration (economic, sociocultural, and environmental) and their adaptation practices in relation to migration intention. Perceptions were collected through a structured questionnaire and measured on a five-point Likert scale. Partial Least Squares Structural Equation Modeling (PLS-SEM) was used to assess the effects of migration drivers and climate change adaptation on migration intention. This study examines whether migration intention is significantly influenced by adaptation to climate change and by the economic, sociocultural, and environmental impacts of migration. It also examines whether adaptation to climate change is influenced by the economic, sociocultural, and environmental impacts of migration. Adaptation to climate change plays a significant mediating role between environmental factors and migration intention, but not between the other drivers (economic and sociocultural) and migration intention. Hence, policy must be revised to develop effective adaptive strategies to mitigate its role in reducing people's intention to migrate.

T022

Neurodevelopment Outcomes of Neonates with Document Sepsis – A Prospective Cohort Study

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Keywords: Neonatal Sepsis, Neuro development, Developmental Delay, Infant Health Low and Middle Income Countries (LMICs)

Neurodevelopmental outcomes in neonates are influenced by factors such as prematurity, congenital and genetic disorders, cardiac conditions, low birth weight and neonatal sepsis. Research from developing countries and low resource settings is limited. In Nepal, no such study assessing the impacts of neonatal sepsis on neurodevelopment has been reported. Understanding these impacts are crucial for early identification of high risk infants and timely interventions. This study aimed to evaluate the effect of neonatal sepsis on neurodevelopmental outcomes, provide context specific evidence to guide clinical care, follow up strategies, rehabilitation programs, and public health policies, ultimately reducing long term morbidity. The study was conducted from February 2023 to November 2025 using a prospective dual-centric cohort design at Paropakar Maternity and Women's Hospital and Siddhi Memorial Hospital. A total of 312 neonates were enrolled through non-probability purposive sampling. Neonates with documented sepsis attending the two study centers were included. Exclusion criteria comprised severe life-threatening conditions unrelated to sepsis, hypoxic-ischemic encephalopathy grade II or III, major congenital, genetic, or metabolic disorders, and extreme prematurity (¡28 weeks) or birth weight ¡800g. Baseline sociodemographic, maternal, and neonatal data were collected. Neurodevelopmental outcomes were assessed at six months and one year after discharge using the Developmental Assessment Scale for Indian Infants (DASII). Statistical analysis was performed using SPSS v 25, applying descriptive statistics, bivariate analysis, and multivariable logistic regression to

estimate prevalence, assess associations, and identify independent predictors of neurodevelopmental delay, with $p < 0.05$ considered statistically significant. Ethical approval was taken Nepal Health Research Council. At six months follow up, neurodevelopmental assessment among 285 infants who survived neonatal sepsis showed motor delay in 20.1%, mental delay in 17.2%, combine motor and mental delay in 11.9%, and any delay in 25.3%. By one year follow up, among 269 infants, motor delay decreased to 13% and mental delay to 5.9%. Motor delay was most common in the neck control cluster at 6 months (36.1%) and in the body control cluster at 1 year (66.9%). Mental delay was highest in the memory cluster at both 6 months (22.5%) and 1 year (68.8%). At 6 months, higher odds of developmental delay were significantly associated with intrauterine growth retardation (aOR = 2.82), gestational diabetes mellitus (aOR = 3.69), postnatal steroid use (aOR = 2.49), and neonatal seizures (aOR = 2.92). At 1 year, antenatal steroid exposure (aOR = 10.08) and neonatal seizures (aOR = 4.17) were the strongest predictors of developmental delay. In this prospective cohort study nearly one-fourth of the neonates who survived documented sepsis demonstrated neurodevelopmental delay at six months. Although the proportion of affected infants declined by one year, one sixth of them continued to have developmental difficulties. Motor delays were most evident in neck and body control, while memory was notably impaired in the mental domain. Intrauterine growth restriction, maternal gestational diabetes, antenatal steroid exposure, and neonatal seizure were significant risk factors for poor outcomes. This study emphasized the importance of early screening, structured follow-up, and timely interventions for septic neonates to improve long term neurodevelopmental outcomes in Nepal.

T023

Study of Statistical Properties of a Non-Monotonic Hazard Model: An Application to Real-Life Data from Engineering and Medical Sciences

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Keywords: Inverted U-shaped hazard; MLE; AIC; KS-test

In this paper we have introduced a new two-parameter continuous distribution with a nonmonotone hazard. Various statistical properties of the distribution, such as MLE, entropy, stochastic ordering, order statistics and ageing intensity have been studied. A simulation study has also been carried out to evaluate the performance of the point and interval estimates derived using maximum likelihood estimation. Furthermore, datasets generating out of real-life situations have also been used to assess the applicability of the distribution. The usefulness of the distribution has been checked by -2loglikelihood, AIC, CAIC and the Kolmogorov-Smirnov test, and we have found that the proposed distribution fits adequately and can be used as an alternative to some well-known distributions.

T024

Exploring Timing of Most Recent Conception: A Data Driven Stochastic Modeling

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Keywords: Stochastic model; Timing of conception; Most recent birth; Maximum likelihood estimation

The conception stage is a critical moment in a woman's life, where insufficient attention may result in negative outcomes such as miscarriage and enduring health consequences that are a crucial driver of a nation's overall growth, necessitating vigilant monitoring of reproductive health outcomes. A significant metric in this context is the timing of birth/conception, which signifies both reproductive patterns and maternal health. The primary aim of this study is to analyze the timing of conception and thus fecundability for the most recent birth among women in Uttar Pradesh, India, utilizing data from the National Family Health Survey (NFHS) and focusing on selected indicators through some proposed stochastic models. This study assesses the waiting time to the most recent closed birth among women utilizing stochastic models such as length-biased exponential distribution (LBED), length-biased exponential gamma distribution (LBEGD) and modified mixed exponential distribution (MMED). The estimate of the parameters has been obtained through the maximum likelihood estimation procedure. The models were utilized on data from Uttar Pradesh and the Varanasi district and thereafter expanded to examine temporal trends over three rounds of the National Family Health Survey. The investigation examined demographic and socio-economic disparities, encompassing women's age, parity, sex of the most recent child, religion, caste, and residential location. The findings demonstrate that the proposed models well fit the data, revealing that over the years, the average waiting time to conception has risen, while fecundability has decreased across the majority of categories. These findings emphasize changing reproductive dynamics and reinforce the necessity for focused policy measures to enhance maternity and child health services.

T025

Ranking Strategic Retail Items by Transaction Quality: Matched Excess Extra-Category Attached Value

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Keywords: Market basket analysis; Transaction quality; Coarsened exact matching; Average treatment effect on the treated; Empirical Bayes shrinkage; Assortment planning; Retail analytics

Merchants protect exclusives and everyday staples on the belief that those SKUs sit in better rest-of-store trips. That belief is rarely turned into an item rank. Pairwise lift cannot do it: the statistic needs a partner SKU, so a unique with no chosen consequent has no score. Ticket averages and blended indexes leave own price in the outcome and mix unlike shopping occasions, so an expensive good or a Saturday stock-up looks like quality by construction.

We take leftover revenue after the item and its subcategory come off the ticket, and compare it with trips that never carried the item but share a store cluster, week of year, weekday, and daypart. Store identity is too fine—cells empty—and basket size is the leftover itself, so neither enters the match. We call the difference excess extra-category attached value (EAV). Uniques are not interchangeable, so noisy estimates shrink toward zero rather than a peer mean. Rank is withheld unless treated volume and match rate clear preset gates. Total excess attached value (TEAV) scales EAV by trip count and must not be summed. A margin analogue (EAM) sits beside EAV and is never averaged with it. Language stays on matched association, not availability incrementality.

A stylized annual table shows why this ordering is built to disagree with sales, basket value, and lift. Lift stays the adjacency tool. The residual rank is a dollar score of rest-of-basket association that does not need a driving item. A live comparative study is left to subsequent work.

T026

Statistical Insight and Human-Centered Data Science: Reimagining Governance, Risk, and Development Pathways in Nepal's Federal Transition

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Keywords: Human-Centered Data Science; Statistical Modeling and Governance; Climate and Risk Analytics; Federalism and Evidence-Based Policy; Ethical and Responsible AI Systems

Nepal's federal transition has unfolded alongside an unprecedented expansion in data availability, computational capacity, and analytical innovation. As governments confront complex challenges from climate vulnerability and public health disparities to administrative fragmentation and uneven development, statistics and data science have become essential tools for shaping responsive and human-centered governance. This paper explores how emerging statistical methodologies, machine learning techniques, and GenAI-driven systems can strengthen Nepal's evidence-based policymaking, while also acknowledging the social, ethical, and institutional realities that shape their application. Drawing on examples from climate risk modeling, public health analytics, environmental monitoring, and local governance reforms, the study highlights how statistical rigor and modern data science can illuminate patterns of vulnerability, support targeted interventions, and enhance transparency across federal, provincial, and local institutions. At the same time, the paper emphasizes the human dimension of data: the lived experiences behind indicators, the inequalities embedded in datasets, and the need for interpretability and accountability in algorithmic decision-making. The analysis argues that Nepal's development trajectory will increasingly depend on its ability to integrate statistical literacy, interoperable data infrastructures, and ethical AI frameworks into everyday governance. Bayesian modeling, spatial analytics, and machine learning offer powerful tools for small-area estimation, climate adaptation planning, and health surveillance, but their effectiveness requires institutional capacity, cross-sector collaboration, and culturally grounded implementation. By situating Nepal's experience within global trends in statistics and data science, the paper proposes a roadmap for strengthening analytical ecosystems that are not only technically robust but also socially responsive. Ultimately, it calls for a data future where statistical insight and human-centered design work together to support equitable development and resilient governance.

T027

Wealth on Paper, Poverty in Pocket: A Latent Class and Ordinal Logit Analysis of Subjective Economic Well-Being in Nepal

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Keywords: Asset-Rich, Cash-Poor; Latent Class Analysis; Subjective Economic Well-Being; Liquidity Vulnerability; Ordinal Logistic Regression

Land and housing values are considered informal proxy indicators of household's wealth in Nepal, but many households in Nepal who have substantial fixed assets also struggle in the process of providing cash for their daily needs. This study examines the complexities of this "asset rich, cash poor" phenomenon, which can be hard to capture with traditional measures of poverty. The study used primary survey data from 207 households to analyze eight indicators of household financial resiliency using a technique called Latent Class Analysis (LCA): housing-status congruence, cash-flow stability, urban cost pressures, discretionary spending capacity, emergency liquidity, mandatory-expense burden, and confidence in assets as crisis insurance. The majority group (58.5%) is the Asset-Rich, Cash-Poor class, in which liquidity is constrained despite their substantial assets, reflecting economic instability; the Deprived class (14.0%) has chronic liquidity problems; the Constrained class (27.5%) has moderately constrained economic conditions. The results of the ordinal logistic regression showed that class membership was highly correlated with the self-assessed economic status. The Deprived and Constrained classes were significantly more likely to consider themselves to be poor than the Cash-Poor class ($p < .001$). Importantly, 97.5% of Asset-Rich, Cash-Poor households self-rated as "Middle" were identified as having asset-based poverty, but lacked liquidity. These findings underscore the fact that asset-based measures can mask important financial vulnerabilities among households that might be considered middle class in Nepal.

T028

Factors Associated with Health-Symptoms Among Agrochemical-Using Farmers in Bharatpur Metropolitan City, Chitwan

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Keywords: agrochemical, knowledge, attitude, safety practice, health-symptoms

This study examined the use of pesticides and chemical fertilizers, knowledge, attitude, safety practices, personal protective equipment (PPE) use, and self-reported health-symptoms among agrochemical-using vegetable farmers in Wards 25, 26, 27, and 28 of Bharatpur Metropolitan City, Chitwan. A cross-sectional survey design was used, and data were collected through face-to-face interviews from 110 eligible vegetable farmers selected using purposive sampling. The data were analyzed using IBM SPSS Statistics. Frequencies and percentages were used for categorical variables, while means and standard deviations were used for composite scores. Pearson's chi-square test of independence was used to examine associations between socio-demographic characteristics, knowledge, attitude, safety practice, and health-symptom level at the 5% significance level. All respondents reported using insecticides and fungicides, while 84.5% reported using herbicides. Urea and DAP were each reported by 99.1% of respondents, and potash by 93.6%. The mean knowledge score was 9.30 ± 2.03, with 48.2% classified as having good knowledge and 51.8% as having poor knowledge. The mean attitude score was 4.91 ± 1.05; 66.4% of respondents had a favourable attitude and 33.6% had a less favourable attitude. The mean safety-practice score was 6.72 ± 3.09, with 50.9% classified as having good safety practices and 49.1% as having poor safety practices. The mean health-symptom score was 4.07 ± 3.22, with 53.6% of respondents classified as having lower symptoms and 46.4% as having higher symptoms. All socio-demographic variables showed no association with health symptoms, as the reported p-values were greater than 0.05. Knowledge was not significantly associated with health-symptoms ($\chi^2 = 3.062$, df=1, $p = .080$), and attitude was also not significantly associated with health-symptoms ($\chi^2 = 1.326$, df=1, $p = .250$). In contrast, safety practice was significantly associated with health-symptoms ($\chi^2 = 24.859$, df=1, $p < .001$). The findings suggest that improving safe agrochemical-handling practices should be a priority in the study area. Because the study was cross-sectional and based partly on self-reported information, the observed associations should not be interpreted as causal relationships.

T029

Seasonality Trends in Child Deaths in India: A Directional Statistical Approach

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Keywords: Child death, Seasonal Variation, Rao's spacing test, Rayleigh test

Seasonal variation in child deaths (deaths of children under five years of age) carries important implications for child health policy, yet identifying when and why deaths cluster across the calendar year requires methods that respect the cyclical nature of monthly data. This study applied circular statistical methods to examine seasonal patterns in the distribution of child deaths in India, analysing data disaggregated by state, year, religious group, and caste category using birth recode data from the National Family Health Survey (NFHS-5), covering the period 2015 to 2020. Rao's spacing test was used as the primary test of seasonal non-uniformity, supported by the Rayleigh test as a secondary measure of unimodal directional concentration. Rao's spacing test confirmed statistically significant seasonal clustering in the distribution of under-five deaths across all nineteen states, all six study years, all three religious groups, and all four caste categories examined. The Rayleigh test reached significance in only five states and two years, a systematic divergence explained by the three to five distinct monthly peaks identified in most analytical units. This multi-peak structure causes directional vector cancellation in the Rayleigh framework, artificially suppressing the mean resultant length and inflating p-values. This divergence itself is an epidemiologically meaningful finding: it confirms that under-five deaths in India are not driven by a single seasonal hazard but by the simultaneous operation of multiple disease pathways on overlapping seasonal calendars. Waterborne and vector-borne diseases including diarrhoea, malaria, and dengue drive monsoon-period peaks (June–September); acute respiratory infections dominate the winter months (December–February); heat stress and early diarrhoeal illness characterise the pre-monsoon period (March–May); and residual vector activity extends into the post-monsoon months (October–November). After accounting for the unequal lengths of calendar seasons, the apparent monsoon dominance widely reported in prior literature was shown to be a methodological artefact rather than a genuine epidemiological signal. Among social groups, Scheduled Tribes carried the highest absolute count of deaths across all seasons, reflecting compounding vulnerabilities of geographic remoteness, nutritional risk, and limited healthcare access. The anomalous pattern observed in 2020 with elevated death counts concentrated in the post-monsoon months is consistent with documented disruptions to child healthcare access and immunisation programmes during the COVID-19 pandemic. These findings indicate that child health programmes concentrated on the monsoon season alone will miss a substantial share of preventable deaths. Year-round service delivery, with targeted intensification during identified peak periods, is more appropriate for most Indian states than seasonally focused campaigns.

T030

Statistical Modelling of Anxiety Level Among Public and Private Students at Higher Secondary Level

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Keywords: Anxiety, Beck Anxiety Inventory, higher secondary students, institutional type, academic stream

Anxiety is an important psychological concern among higher secondary students who may experience academic demands, examination-related concerns, career expectations, and psychosocial pressures. This study aimed to assess anxiety levels and statistically model factors associated with anxiety among public and private higher secondary students of Bharatpur Metropolitan City, Chitwan. A quantitative analytical cross-sectional study was conducted among 300 Grade 11 and Grade 12 students from Science and Management streams selected using multistage sampling. Anxiety was assessed using the Beck Anxiety Inventory (BAI). Descriptive statistics, independent-samples t-test, two-way analysis of variance, bivariate analysis, and multiple linear regression were used for statistical analysis. Separate regression models were developed for public and private colleges. The mean BAI score was 19.22 ± 11.35, ranging from 0 to 63. About 60.6% of students experienced moderate-to-severe anxiety. Public-school students had significantly higher mean BAI scores than private-school students (21.90 ± 11.32 vs. 17.06 ± 10.94, $p = .001$), whereas the difference by academic stream was not significant ($p = 0.647$). Two-way ANOVA showed significant effects of college type ($p = 0.001$) and its interaction with academic stream ($p = 0.028$). The regression model identified fear of result, sleep duration, and female gender as significant predictors of anxiety in public schools, while birth order, fear of result, family-related factors, sleep duration, and perceived course difficulty were associated with anxiety in private schools. The findings demonstrate that anxiety is common and is associated with multiple academic, familial, and psychosocial factors, with patterns varying by institutional type.

T031

An Extended Technology Acceptance Model (TAM) For Behavioral Intention To Adopt Digital Payment: A Study Of University Students

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Keywords: Digital payment, TAM, UTAUT, behavioral intention, university students, PLS-SEM

The rapid expansion of digital payment systems in Nepal has increased the need to understand the factors that influence their adoption intentions, particularly among university students. This study examines the determinants of university students' behavioral intention to adopt digital payment systems using an extended Technology Acceptance Model (TAM) incorporating trust, perceived risk, facilitating conditions, and subjective norms. Data were collected through a structured questionnaire from 401 university students in Nepal and analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM). The results show that 7 out of 11 proposed hypotheses were supported. Perceived ease of use significantly influences perceived usefulness and attitude toward using digital payments, while perceived usefulness also positively influences attitude. Attitude toward using digital payments has the strongest direct effect on behavioral intention. Trust positively influences perceived usefulness, perceived risk negatively affects trust, and facilitating conditions positively influence behavioral intention. However, perceived usefulness, trust, perceived risk, and subjective norms do not have significant direct effects on behavioral intention. Overall, the findings indicate that digital payment adoption among university students is a multidimensional process, with attitude, ease of use, and facilitating conditions playing particularly important roles, while trust and perceived risk primarily operate through users' perceptions rather than directly determining intention. Thus, the extended TAM provides a meaningful and useful framework for understanding digital payment adoption among university students in Nepal.

T032

Factors Affecting Child Nutritional Status In Nepal: A Secondary Data Analysis Using Nepal Demographic & Health Survey (NDHS) 2022

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Keywords: Child malnutrition, stunting, wasting, underweight, NDHS 2022, Nepal

Child malnutrition remains a critical public health challenge in Nepal, with persistent disparities across socioeconomic and geographic groups. This study examines factors associated with stunting, wasting, and underweight among children aged 0–59 months in Nepal using child, maternal, household, and community-level characteristics. Data were drawn from the Nepal Demographic and Health Survey (NDHS) 2022, comprising a sample of 2,202 children with complete anthropometric measurements.

Nutritional status was assessed using three indicators—stunting (height-for-age), wasting (weight-for-height), and underweight (weight-for-age)—based on WHO Child Growth Standards. Descriptive statistics and Chi-square tests were employed to examine associations between selected factors and child nutritional outcomes. The findings indicate that child age, birth order, size at birth, maternal education, maternal age, maternal employment status, household wealth, sanitation facility, place of residence, and province were significantly associated with one or more forms of malnutrition.

Stunting and underweight were highest among children aged 24–35 months, whereas wasting was most prevalent among children aged 6–11 months. Children from poorer households, those with unimproved sanitation, and those living in rural areas experienced higher levels of malnutrition. Significant provincial disparities were also observed, with Karnali reporting the highest stunting prevalence and Madhesh reporting the highest wasting and underweight prevalence. In contrast, sex of the child and source of drinking water did not show statistically significant associations with any nutritional outcome. Overall, the findings underscore the multidimensional nature of child malnutrition in Nepal and highlight the importance of maternal education, improved household socioeconomic conditions and sanitation, and geographically targeted interventions in addressing child undernutrition.

T033

Determinants Of Graduate Students' Satisfaction With Facilities And Services At The Central Departments Of IoST, Tribhuvan University

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Keywords: Student Satisfaction, SERVPERF Model, Higher Education, Graduate Students, PLS-SEM

This Study argues that the existing frameworks of student satisfaction in higher education, while being widely applied, have a greater emphasis on material facilities and structural resources with little consideration of relational and experiential aspects in shaping student perceptions. This study uses the adapted SERVPERF model and a census-based sample of 321 third-semester master's degree students in the 13 Central Departments of the Institute of Science and Technology (IoST). The study uses Partial Least Squares Structural Equation Modeling (PLS-SEM) to assess the impact of six dimensions of service quality: Tangible, Assurance, Reliability, Responsiveness, Empathy, and Extracurricular Activities, on overall student satisfaction.

The results show that the only statistically significant predictor of student satisfaction is Empathy, which includes such sub-elements as support of the faculty, fairness, accessibility, and individual academic care. In contrast, the conventional dimensions, such as Tangibles, Reliability, and Responsiveness, do not differ significantly, indicating that in situations where infrastructural constraints are present, students are oriented toward valuable human interaction rather than physical or administrative facilities.

The findings confirm that student satisfaction is a multi-dimensional phenomenon, yet Nepalese higher education is affected less by infrastructural adequacy and more by considerable academic support, teacher-student interaction, and relational caring. Thus, this study contributes to the broader literature by challenging facility-centered models of satisfaction and illustrating the usefulness of relational quality for explaining and improving student experience.

T034

Analyzing Factor Associated With Most Recent Closed Birth Interval Using Classification And Regression Tree Modelling Approach

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Keywords: Most recent closed birth interval, Multinomial regression, Classification and Regression Tree (CART)

Background: The study on most recent closed birth interval is of immense importance for a country like India. Among the several birth intervals, the most recent closed birth interval is determined to be more essential than others to know the recent fertility pattern. The present study was undertaken to examine the effect of several parameters like age at marriage, age at first birth, religion, caste, parity, sex of the kid, education, residence, contraceptive usage, and wealth index on Most recent closed birth interval.

Methodology: A multinomial logistic regression and Classification and Regression Tree modelling technique (CART) survival tree method are employed to accurately examine the effect of specified factors on MRCBI.

Results: Multinomial logistic regression identified age at marriage, age at first birth, present age, and survival status of prior kid as the key factors of birth interval. Women who married and started childbearing at older ages were more likely to have short birth intervals. Older women were more likely to have long birth intervals. Survival of preceding kid lowered the chances of short birth intervals and raised the chances of lengthy birth intervals. The CART model confirmed these findings and identified maternal age, reproductive history, and child survival status as the most relevant determinants of birth interval.

Conclusion: The combination of multinomial logistic regression and CART allows for a full knowledge of birth interval dynamics and can help to identify women at risk of poor birth spacing, which can in turn enable targeted maternal and child health interventions and family planning initiatives.

T035

Statistical Distribution Of Annual Tree Growth In The Nepal Himalaya

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Keywords: Annual ring width, Dendrochronology, Gamma distribution, Himalayan forests, Lognormal distribution, Statistical modeling, Tree growth

Annual tree-ring width is a fundamental measure of radial growth in dendrochronology, but its statistical distribution should be examined before choosing a growth or climate-response model. Forests provide various ecosystem services, including fuel wood and timber, among others. Nepalese forests are affected by climate change, but there are several gaps in knowledge about the quantitative growth of many tree species in Nepal. Dendrochronology has wider applications in multiple sectors, including quantitative time-series analysis. Therefore, this study was carried out to analyze quantitative growth and its distribution in the Nepal Himalaya. By collecting tree-ring data from 63 sites across Nepal, this study analyzed the growth of 12 tree species having different age ranges from about 50 years to 1100 years. The average age of the sampled trees was 70 years, and the average growth was 2.07 mm/year. The distributional analysis of tree growth showed that the growth was right-skewed and not normally distributed. This study demonstrates a distributional analysis using a synthetic example representing 20 trees observed annually from 1996 to 2020 ($n = 500$). Annual ring widths were generated to resemble positive, moderately right-skewed growth measurements from Himalayan conifers. Descriptive statistics and four continuous probability distributions—Normal, Lognormal, Gamma, and Weibull—were compared using maximum likelihood, Akaike's information criterion (AIC), and the Kolmogorov–Smirnov (KS) statistic. The example dataset had a mean annual ring width of 2.07 mm, standard deviation of 0.97 mm, coefficient of variation of 0.47, and skewness of 1.00. Among the candidate models, the Gamma distribution produced the lowest AIC (1297.4), indicating the best relative fit within this illustrative comparison. The results emphasize that annual growth is non-negative and may be asymmetric, so distributional diagnostics should precede regression, time-series, or hierarchical modeling. In actual Nepal Himalaya research, the synthetic data should be replaced by crossdated field measurements, followed by appropriate detrending/standardization and assessment of temporal dependence.The analysis provides a reproducible statistical framework for evaluating the distribution of annual tree growth and for selecting probability models suitable for subsequent climate-growth modeling.

T036

An Investigation Of Reproductive Complications And Intrauterine Device (IUD) Users In Uttar Pradesh, India

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Keywords: Reproductive complications, IUD users, NFHS-5, logistic regression

Objectives: Intrauterine devices (IUDs) are an invasive method of modern family planning. It is evident that the use of IUDs leads to reproductive complications such as vaginal discharge, irregular menstruation, bad odour, etc. The occurrence of reproductive complications varies across different demographic and socioeconomic groups. The purpose of this study is to examine the sociodemographic determinants of reproductive complications and use of IUDs using NFHS-5 data.

Materials and Methods: The information was collected from women surveyed in the National Family Health Survey (NFHS-5) to evaluate reproductive complications. The study used bivariate analysis and logistic regression to examine the individual and household characteristics—including maternal age, place of residence, caste, religion, wealth index, parity, and educational attainment—associated with reproductive complications among IUD users compared to the total population.

Results: Findings reveal that reproductive complications are heavily concentrated among specific vulnerable subgroups. Younger women under the age of 25 and primiparous women using IUDs experience substantially higher rates of reproductive complications compared to older and higher-parity cohorts. Socioeconomic and geographic factors also serve as strong predictors: women residing in rural areas, those belonging to marginalised castes (SC/ST), and those from the poorest wealth quintiles face a notably higher burden of complications than their urban, general-caste, and wealthier counterparts. Higher reported complications among educated women further indicate the role of symptom awareness and health literacy in reporting adverse reproductive events.

Conclusions: Results reveal that reproductive complications among IUD users are significantly influenced by maternal age, parity, wealth index, caste, place of residence, religion, and educational level. This study specifies the need for adequate pre-insertion clinical counselling, skilled provider training in rural and resource-limited settings, and routine post-insertion follow-up to improve contraceptive safety and reproductive healthcare.

T037

People's Knowledge Of Climate Change's Impact On Tourism In The Annapurna Conservation Area, Nepal, By Developing A Climate Change Knowledge Index

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Keywords: Climate change, knowledge index, tourism, impact, Nepal

Nepal is experiencing significant environmental impacts from rising temperatures, glacial melt, and increasingly frequent extreme weather events. These changes have adversely affected public health, livelihoods, ecological systems, and the tourism sector. This study used survey data from 490 households in the Annapurna Conservation Area to assess knowledge of climate change impacts on tourism. The findings reveal that climate change awareness varies significantly by geographic location and demographic characteristics and is closely associated with age, education level, and place of residence. Urban residents exhibit notably higher awareness than their rural counterparts, and individuals with higher education demonstrate substantially greater knowledge than those with only basic education. Interestingly, literate individuals also show significantly more awareness than those with basic education. The composite climate change knowledge index yields a mean score of 62.73, indicating a moderate-to-high level of awareness among participants. While the majority possess satisfactory knowledge, approximately 42% of the population remains at low or moderate levels. These results suggest that public understanding is largely derived from experiential observations rather than formal scientific sources.

T038

Forecasting Nepal's Nominal Gross Domestic Product Using The Box–Jenkins ARIMA Methodology

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Keywords: ARIMA, Box–Jenkins methodology, GDP forecasting, Nepal, Time series analysis, Nominal GDP

Gross domestic product (GDP) is a key indicator of economic activity and provides an important basis for macroeconomic planning, fiscal management, investment decisions, and public resource allocation. This study models and forecasts Nepal's annual nominal GDP using the Box–Jenkins autoregressive integrated moving average (ARIMA) methodology. Annual nominal GDP data at current prices for the fiscal years 1990/91 to 2024/25, comprising 35 observations and expressed in million Nepalese rupees, were analyzed using R. The GDP series was logarithmically transformed, and stationarity was assessed using graphical analysis, autocorrelation and partial autocorrelation functions, the Augmented Dickey–Fuller (ADF) test, and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. Six pre-specified ARIMA models were estimated using maximum likelihood and compared using AIC, AICc, BIC, log-likelihood, RMSE, MAD, and residual Ljung–Box diagnostics. The results indicated that the GDP series was non-stationary in its original and logarithmic forms, while first differencing of the log-transformed series produced mixed stationarity evidence, with the KPSS test supporting stationarity. Among the pre-specified candidate models, ARIMA(1,1,2) achieved the lowest AIC and AICc and showed no significant residual autocorrelation. However, its autoregressive coefficient was estimated at the boundary of the parameter space, indicating a potential near-unit-root estimation issue. The automated model-selection procedure identified ARIMA(0,1,0) with drift as a more parsimonious specification with lower information criteria. Forecasts from ARIMA(1,1,2) indicate that Nepal's nominal GDP is expected to increase from approximately NPR 6.353 trillion in 2025/26 to NPR 7.908 trillion in 2029/30, although prediction intervals widen substantially with the forecast horizon. The findings demonstrate the usefulness of ARIMA as a transparent benchmark for Nepal's nominal GDP forecasting while highlighting the importance of model comparison, parsimony, and explicit treatment of forecast uncertainty. Future research should incorporate out-of-sample validation and compare univariate ARIMA with multivariate, machine-learning, and hybrid forecasting approaches.

T039

Awareness Regarding Autism Among Parents Attending A Teaching Hospital In Lalitpur, Nepal: A Cross-Sectional Study

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Keywords: Autism Spectrum Disorder, Awareness, Parents, Teaching hospital

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition affecting social communication, interaction, and behavior, which is increasing worldwide. Awareness among parents is crucial as early detection and timely intervention can significantly improve developmental outcomes. Unfortunately, there are limited studies on parental awareness of ASD among hospital-based populations in Nepal. This study aimed to assess awareness regarding autism among parents attending a teaching hospital in Lalitpur and to examine its association with sociodemographic characteristics. A cross-sectional study was conducted among 98 parents attending the teaching hospital in Lalitpur, selected through non-probability convenience sampling. Data were collected using a self-developed structured questionnaire assessing awareness regarding autism, including signs and symptoms, risk factors, and management. Each correct response was assigned one point, and awareness scores were calculated as the sum of correct responses. Descriptive statistics and appropriate inferential statistical tests were used to analyze the data. Statistical significance was considered at $p < .05$. Among the 98 parents, 4 (4.1%) reported a family history of autism. The mean overall autism awareness score was 14.90 ± 4.42. The mean scores for awareness regarding signs and symptoms, risk factors, and management were 6.77 ± 2.78, 3.42 ± 1.67, and 4.04 ± 1.15, respectively. Overall awareness scores were approximately normally distributed (Kolmogorov–Smirnov Z $= 1.28$, $p = 0.060$). There was no statistically significant difference in overall awareness scores according to sex of parents ($p = .616$) or family history of autism ($p = .462$), and age was not significantly correlated with overall awareness score ($r = .164$, $p = .107$). Similarly, no statistically significant difference was observed according to occupation, $F(6, 91) = 1.83$, $p = .102$. However, overall awareness differed significantly according to educational level, $F(3, 94) = 8.02$, $p < .001$. The study found variation in autism awareness among parents, with educational level significantly associated with overall awareness. Strengthening autism-related awareness among parents, particularly those with lower educational attainment, may support earlier recognition of signs and symptoms and appropriate management.

T040

Computational Analysis Of The Effects Of Stenosis Height And Width On Two-Layered Blood Flow Through An Artery With Bell-Shaped Stenosis

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Keywords: Bell-shaped stenosis, stenosis height, stenosis width, computational analysis, two-layered blood flow, wall shear stress, pressure drop, volumetric flow rate

This study presents a computational investigation of two-layered blood flow through an artery with bell-shaped stenosis, with particular emphasis on the effects of stenosis height and stenosis width on key hemodynamic characteristics. The arterial lumen is modeled as a two-layer system consisting of a peripheral plasma layer and a central core region, and the governing Navier–Stokes equations are formulated in cylindrical polar coordinates under the assumptions of incompressible, Newtonian, axisymmetric, and fully developed flow. Analytical expressions are obtained for the velocity distribution, volumetric flow rate, wall shear stress, pressure gradient, and pressure drop in the stenotic region.

A systematic computational parametric analysis is then performed by varying the stenosis height and width independently and in combination to quantify their influence on blood-flow behavior. Numerical integration is employed to evaluate the pressure drop ratio and shear-stress characteristics in both the core and peripheral layers. The computational results demonstrate how increasing stenosis height intensifies the restriction of the flow passage, producing significant changes in velocity distribution, wall shear stress, volumetric flow rate, and pressure loss. The stenosis width is also found to play an important role by controlling the axial extent over which these hemodynamic changes occur. The combined influence of stenosis height and width provides a more comprehensive description of the severity and spatial characteristics of stenotic flow than consideration of either geometric parameter alone.

The study provides a quantitative computational framework for assessing the hemodynamic consequences of different stenosis geometries and contributes to the mathematical modeling of blood flow in stenosed arteries. The findings may be useful for understanding the relationship between stenosis geometry and clinically relevant flow parameters in cardiovascular biomechanics.

T041

Performance Evaluation Of Machine Learning Classification Algorithms For Infectious Disease Prediction Using Inpatient Records From Nepalese Hospitals

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Keywords: Dengue, Disease prediction, Kala-azar, Scrub typhus, Supervised learning

Accurate forecasting of infectious diseases is crucial to better clinical decision-making and lower disease burden in resource-poor healthcare environments. In this study, various machine learning (ML) classification algorithms were used to predict three diseases: dengue, scrub typhus and kala-azar from inpatient records of seven hundred nine from three different hospitals in Nepal. Routine laboratory parameters and demographic parameters were chosen as predictor variables, and having a disease as a dependent variable. In Python, five supervised ML algorithms: Random Forest (RF), Logistic Regression (LR), Artificial Neural Network (ANN), Support Vector Machine (SVM), and Decision Tree (DT) were developed, with the data being split into an 80:20 train–test ratio, and randomization is done for splitting data. The accuracy, precision, recall, F1-score, and area under the receiver (AUC) were used to assess model performance.

ANN achieved the best predictive performance among the dengue cases with an accuracy of 73.9% and an F1-score of 0.82. For the scrub typhus, ANN had the highest accuracy value of 76.8% and F1 score (0.63) while SVM had the highest AUC value of 0.80. For the prediction of kala-azar, ANN exhibited the highest discriminative ability with an AUC of 0.95 and satisfactory accuracy (96.5%) while RF displayed the highest overall accuracy (97.2%). In general, ANN performed best in both regards, being the most balanced and reliable classifier for the three infectious diseases. The findings underscore the possibility of using ML-based clinical decision support systems in the early prediction of infectious diseases in hospitals in Nepal, and how normal laboratory and demographic information could be combined with the potential of routine information in predictive healthcare analytics.

T042

Distributional Pattern Of Labor Productivity In Manufacturing Industries Of Nepal

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Keywords: Labor productivity, Distributional pattern, GB2 distribution, Nepal, Manufacturing Industries

The distributional properties of labor productivity are frequently assumed to follow a lognormal form, consistent with Gibrat's law of proportionate industry growth (Ishikawa et al., 2022). However, this assumption is seldom empirically verified before its application in regression-based productivity research. This study systematically evaluates alternative distributional specifications to identify the model that best represents industry-level labor productivity in Nepal's manufacturing industries. Establishing the correct distributional form is a critical prerequisite for robust inference regarding productivity determinants.

Study data from 480 manufacturing industries were analyzed, with five candidate distributions lognormal, gamma, log-logistic, log-Student's t, and the four-parameter Generalized Beta of the Second Kind (GB2) estimated using maximum likelihood methods. Model adequacy was evaluated using the Akaike and Bayesian Information Criteria, supplemented by graphical diagnostics such as empirical-versus-fitted density overlays and quantile-quantile plots. The observed productivity data demonstrated substantial skewness (9.50) and excess kurtosis (114.39). Comparative analysis revealed that the GB2 distribution provided the best fit (log-likelihood = $-7,043.98$; AIC = 14,095.95; BIC = 14,112.65), followed by the log-Student's t ($\hat{\nu} \approx 2.87$; AIC = 14,160.19) and log-logistic (AIC = 14,166.18) distributions. All three substantially outperformed the conventional lognormal specification (AIC = 14,195.66), whereas the gamma distribution exhibited the poorest fit (AIC = 14,471.59), failing to capture even the central mode.

The findings demonstrate that labor productivity in the Nepali manufacturing industry exhibits genuinely heavy-tailed behavior rather than simple right skewness. This challenges the prevailing reliance on lognormal assumptions and underscores the necessity of adopting more flexible likelihood specifications in subsequent econometric modeling, particularly within Bayesian frameworks for productivity analysis.

T043

Parameter-Efficient Domain Adaptation And Preference Alignment Of An Open Large Language Model For Nepal's Electricity-Sector Regulatory Corpus

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Keywords: Large language models, low-resource languages, Nepali NLP, domain adaptation, LoRA, direct preference optimization, optical character recognition, regulatory text

Large language models underperform on low-resource, domain-specific text such as Nepal's electricity-sector legal and regulatory documents, which are bilingual and distributed as image-only PDFs with no machine-readable text layer. We present NEA-BOT, an end-to-end pipeline that adapts an open 4.7-billion-parameter model (Gemma 4 E2B, reimplemented from scratch in PyTorch) to this domain under academic compute constraints.

We apply optical character recognition to 107 Nepal Electricity Authority and statutory PDFs using a Nepali-tuned Tesseract model, then construct a cleaned bilingual corpus of 4,845 passages totalling 3.70 million tokens (approximately 40% Devanagari and 60% English), from which we synthesise 9,632 grounded instruction pairs and 9,615 preference pairs. Three sequential parameter-efficient stages—continued pretraining (CPT), supervised instruction fine-tuning (SFT), and Direct Preference Optimization (DPO)—each train only 25.3 million low-rank adaptation (LoRA) parameters, representing 0.54% of the model, on a single commodity GPU in roughly 70 minutes in total.

On deterministic held-out evaluation, CPT substantially improves domain modeling, reducing perplexity from 9,194.8 to 7.8. SFT produces the largest gains in answer quality, increasing token-F1 from 0.139 to 0.546 and ROUGE-L from 0.119 to 0.525. DPO substantially improves preference alignment, increasing preference accuracy from 0.095 to 0.850. However, this alignment improvement is accompanied by a decline in language-modeling and answer-generation performance, with perplexity increasing to 24.4 and token-F1 decreasing to 0.233. These results reveal a measurable alignment tax and demonstrate the importance of stage-wise evaluation rather than relying on a single performance metric.

The work demonstrates that credible domain adaptation and preference alignment for low-resource, bilingual regulatory text can be achieved on commodity hardware. It contributes a reproducible training pipeline and stage-wise evaluation framework for public-sector and regulatory natural language processing in Nepal.

T044

High-Risk Fertility Behavior and Its Associated Factors among Women with Recent Births in Nepal: Evidence from the 2022 Nepal Demographic and Health Survey

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Keywords: High-risk fertility behavior, Maternal and Child Health, Nepal

High-risk fertility behavior (HRFB) is an important public health concern with adverse implications for maternal and child health. This study provides an updated national assessment of HRFB among women with a recent birth in Nepal using data from the 2022 Nepal Demographic and Health Survey. A weighted sample of 4,377 women was analyzed, and multivariable binary logistic regression was used to identify factors associated with HRFB. Overall, 32.3% (95% CI: 30.9–33.7%) of women experienced at least one HRFB. Higher educational attainment of women was strongly associated with lower odds of HRFB, with adjusted odds ratios of 0.51, 0.22, and 0.16 for basic, secondary, and higher education, respectively. Husband's basic and secondary education was also associated with reduced odds. Muslim women had higher odds of HRFB than Hindu women (AOR: 1.52, 95% CI: 1.09–2.11), while women from poor and middle-income households had higher odds than those from rich households (AOR: 1.88 and 1.38, respectively). Women whose healthcare decisions were made by someone else also had higher odds of HRFB (AOR: 1.26, 95% CI: 1.02–1.54). These findings indicate that HRFB affects nearly one-third of women with a recent birth in Nepal and is associated with education, household wealth, religion, and women's decision-making autonomy. Strengthening reproductive health programs, improving female and partner education, and promoting women's empowerment may help reduce high-risk fertility behavior in Nepal.

T045

A Hybrid CNN–Random Forest Approach for Plant Disease Detection

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Keywords: Plant disease detection, CNN, Random Forest, image classification, precision agriculture

Plant diseases can significantly reduce crop quality and production when they are not identified at an early stage. This study presents a web-based Plant Disease Detection System (PDDS) that identifies plant diseases from leaf images and provides users with information about their causes, prevention, and treatment. The proposed system follows a hybrid learning approach in which a convolutional neural network (CNN) extracts visual features, such as color, texture, shape, and disease patterns, while a random forest classifier assigns the extracted feature vector to a particular disease class. The model was developed using 29,245 RGB leaf images, each with a dimension of 256×256 pixels. The dataset consists of 22 healthy and diseased classes associated with apple, grape, corn, and tomato plants. Exploratory data analysis showed a considerable difference in the number of images available for each class, indicating class imbalance within the dataset. The model was evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, as well as ROC analysis. Accuracy was calculated as $A = (TP + TN)/(TP + TN + FP + FN)$, whereas the relationship between precision (P) and recall (R) was measured using $F_1 = 2PR/(P + R)$. The trained model achieved approximately 98% validation accuracy, while precision, recall, and F1-score each reached 97.78%. The findings indicate that CNN-based feature extraction combined with random forest classification can effectively distinguish most plant disease patterns. However, diseases with visually similar or overlapping symptoms remain comparatively difficult to classify. The prediction model was integrated into a Django-based web application, while MongoDB Atlas was used to manage user, plant, and disease-related information. Functional, system, responsiveness, and usability tests confirmed that the main system modules performed as expected. The developed system offers an accessible method for preliminary plant disease screening, although its reliability is influenced by leaf-image quality, background conditions, and the range of crops represented in the training dataset.

T046

Statistical Analysis of Knowledge and Practice of Sexual and Reproductive Health Rights Among Married Women of Reproductive Age at Lamahi Municipality, Dang, Nepal

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Keywords: SRHR, Logistic Regression, MWRA, Underage Marriage, Omnibus Test, Hosmer–Lemeshow, ROC-AUC, Nepal

Sexual and Reproductive Health Rights (SRHR) encompass bodily autonomy, safe sexual lives, and free reproductive choices, including decisions on if, when, and how often to reproduce. Despite constitutional protections in Nepal, awareness and utilization remain low among married women of reproductive age (MWRA). This study assessed the levels of SRHR knowledge and practice among MWRA residing in Lamahi Municipality, Dang, and identified associated socio-demographic and economic factors. A cross-sectional study was conducted among 403 married women selected through a two-stage stratified random sampling technique. Data were collected through structured face-to-face interviews and analyzed using R software version 4.3.3. Descriptive statistics were used for univariate analysis, while Chi-square and Fisher's exact tests were applied for bivariate analysis. Variables significant at the bivariate analysis were entered into multivariable binary logistic regression models to identify independent predictors of SRHR knowledge and practice. Model adequacy was assessed using the Omnibus test, Hosmer–Lemeshow test, classification accuracy, and ROC-AUC analysis. The study revealed that 45.7% of respondents had adequate knowledge and 42.9% demonstrated good practice of SRHR. More than half (55.10%) of the respondents were married before the legal age of 20 years. Multivariable logistic regression analysis showed that respondents with secondary or higher education had significantly higher odds of adequate SRHR knowledge (AOR = 5.068, 95% CI: 2.657–9.665, $p < 0.001$) and good SRHR practice (AOR = 2.813, 95% CI: 1.552–5.098, $p = 0.001$). Husband's education was also significantly associated with both knowledge (AOR = 2.146, $p = 0.013$) and practice (AOR = 3.047, $p < 0.001$). High family income significantly increased the likelihood of adequate knowledge and good practice, while women aged 35–49 years had lower odds of good practice (AOR = 0.284, $p = 0.006$). Marriage at or after 20 years significantly improved SRHR knowledge (AOR = 1.894, $p = 0.031$). The models demonstrated good fit and predictive performance, with ROC-AUC values of 0.885 for knowledge and 0.846 for practice.

Knowledge and practice regarding SRHR among MWRA were found to be inadequate and poor, strongly influenced by educational attainment, economic status, and age at marriage identified in the study. Strengthening women's education, promoting economic empowerment, delaying early marriage, and increasing male involvement are essential to improving SRHR outcomes in the study area.

T047

Unveiling Nepal's Data Revolution: The Role of Statistical Decision-Making in Advancing the Sustainable Development Goals

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Keywords: Data Revolution, Statistical Intelligence, Evidence-Based Decision-Making, Sustainable Development Goals, SDG Monitoring, Nepal, Data Governance

The rapid expansion of data systems is reshaping how societies understand development and make public decisions. For Nepal, where evidence-based governance is increasingly essential for achieving the Sustainable Development Goals (SDGs), transforming growing volumes of data into reliable statistical intelligence remains a critical challenge. Nepal's 2024 SDG review reported data availability for 79.1% of 301 indicators, while highlighting continuing gaps in data coverage and disaggregation (United Nations, 2024). This study examines the data-to-decision nexus by exploring how statistical evidence can strengthen SDG planning, monitoring, and policy decisions in Nepal. Using a mixed-method design, the study integrates SDG indicators, national statistical datasets, and policy evidence through descriptive statistics, correlation, regression modelling, trend analysis, and robustness diagnostics. Results are expected to show that stronger statistical capacity, data quality, accessibility, and evidence-based decision-making improve SDG monitoring and policy responsiveness. The study concludes that Nepal's data revolution can advance sustainable development only when credible data are transformed into actionable decisions.

T048

Neuromorphic Computing and Spiking Neural Networks: A Brain-Inspired Pathway Toward Artificial General Intelligence

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Keywords: Neuromorphic computing, spiking neural networks, spike-timing-dependent plasticity, artificial general intelligence, von Neumann bottleneck, in-memory computing, surrogate gradient learning, neuromorphic hardware

Contemporary deep learning systems achieve strong performance on narrow benchmarks but remain constrained by the von Neumann architecture, which separates memory and processing and relies heavily on dense floating-point operations and global backpropagation. This study examines neuromorphic computing and spiking neural networks (SNNs) as a potentially more energy-efficient computational substrate for artificial general intelligence (AGI), focusing on event-driven spike representations, in-memory computing, and biologically grounded local learning rules. The research will combine simulation and hardware validation using spiking architectures such as leaky integrate-and-fire and adaptive neuron models implemented with open-source frameworks, including snnTorch and Intel's Lava. Surrogate-gradient training will be compared with spike-timing-dependent plasticity (STDP) and related local learning approaches using event-based and static benchmark datasets. Trained networks will subsequently be evaluated on neuromorphic platforms, including Intel's Loihi 2 and BrainChip's Akida, with emphasis on energy consumption, inference latency, and on-chip learning performance relative to GPU-based approaches. The study is expected to characterize the trade-off between biological plausibility and task accuracy, identify key algorithmic and hardware barriers to scalable SNNs, and propose a candidate local learning rule for improving their performance on complex reasoning tasks. The findings aim to contribute to the development of energy-efficient and continually learning systems for edge robotics, autonomous sensing, and always-on biomedical applications, while positioning neuromorphic computing within the broader landscape of AGI research.

T049

A Comparison of Open-source and Proprietary Analytics in Innovative Statistical Applications, Spanning Statistical Methods and Software to Intelligent Data Analysis.

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Keywords: Open-source analytics, Proprietary analytics, Statistical software, Intelligent data analysis, Statistical computing, Machine learning, Reproducibility, Data science

The rapid evolution of statistical computing has expanded the range of tools available for statistical analysis, data visualization, predictive modelling, and intelligent data-driven decision-making. Open-source platforms such as R and Python provide flexible, transparent, and extensible environments, whereas proprietary software such as SAS, SPSS, and Stata offers integrated workflows, standardized procedures, and extensive technical support. This study presents a comparative assessment of open-source and proprietary analytics environments across innovative statistical applications, covering statistical methods, software capabilities, computational efficiency, reproducibility, visualization, machine learning, and intelligent data analysis. The comparison considers their strengths and limitations in handling conventional statistical procedures as well as emerging applications involving large and complex datasets. Particular attention is given to accessibility, scalability, interoperability, automation, user-friendliness, and the availability of advanced analytical methods. The study also examines how software choice influences reproducibility, collaboration, and the practical implementation of modern statistical techniques. The findings are expected to show that neither approach is universally superior; rather, their relative advantages depend on the analytical task, user expertise, computational requirements, institutional resources, and desired level of customization. Open-source analytics offer substantial advantages in flexibility, transparency, and extensibility, while proprietary platforms remain valuable for standardized workflows, enterprise-level applications, and structured technical support. The study highlights the importance of selecting analytical tools based on methodological requirements rather than software preference and provides practical guidance for researchers, statisticians, educators, and organizations adopting modern statistical and intelligent data-analysis approaches.

T050

Beyond Distance-Decay: A Graph Attention Network for Interpretable District-Interaction Analysis in Nepal's Sub-National Education System

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Keywords: Graph Attention Networks, Spatial Interdependence, Education Equity, Sub-National Policy Analysis, Interpretable Machine Learning, Nepal

Sub-national education research in Nepal has traditionally treated the 77 districts as spatially independent units, or at best modeled their interdependence through fixed-weight spatial models (e.g., Geographically Weighted Regression, Spatial Lag Models) that assume influence decays purely with geographic distance. This study introduces a multi-relational Graph Attention Network (GAT)—to our knowledge, the first application of Graph Neural Networks to sub-national education policy in a developing-country context—that instead learns which inter-district relationships matter for predicting literacy outcomes. A district graph is constructed from three interaction channels: spatial adjacency, socioeconomic similarity, and institutional similarity (teacher quality, student-teacher ratio, infrastructure), yielding 3,750 edges across 77 districts and 7 provinces. A 3-head GAT is trained to predict district literacy from 10 socioeconomic and institutional covariates, with attention weights recovered as an interpretable by-product of the message-passing process. Contrary to the classical distance-decay assumption, learned attention is dominated by socioeconomic and institutional similarity channels rather than pure geographic adjacency, and identifies a small set of high-influence "bridge" districts (e.g., Syangja, Kavrepalanchok, Rasuwa) whose attention connects otherwise distant provinces. However, leave-one-province-out validation shows all models, including linear baselines, generalize poorly ($R^2 < 0$) at this district count, exposing a statistical challenge specific to small-N sub-national graphs that the field has not yet addressed. We argue that attention-based interpretability, rather than raw predictive accuracy, is the more actionable output of graph learning at this scale, and propose attention-weighted "bridge" districts as targets for policy interventions designed to maximize network spillovers.

T051

Statistics in the Context of Modern Data Science and AI: A Case Study in Explainable Policy Analytics

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[1]

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Keywords: Statistics, Data Science, Artificial Intelligence, Explainable AI, Model Validation, Policy Analytics

As data science and artificial intelligence continue to reshape research and decision-making, classical statistical methods remain essential for ensuring that data-driven conclusions are valid, interpretable, and trustworthy. This work explores how core statistical techniques—hypothesis testing, model comparison, and structure estimation—can be integrated with modern machine learning and explainable AI to support real-world decision-making. Using district-level educational data as a case study, we combine a data-driven structure-learning method with SHAP-based feature attribution and statistical model comparison (BIC, bootstrap resampling) to evaluate which factors most influence outcomes and how confidently those relationships can be trusted. The results highlight a broader lesson for applied statistics in the AI era: data-driven and theory-driven approaches do not always agree, and rigorous statistical validation remains critical when machine learning is used for policy and decision support. This case study illustrates how statistics and data science, working together, can produce more transparent and reliable insights than either approach alone—a theme central to the emerging applications of statistics in modern research.

T052

Forecasting the NEPSE Index using ARIMA Modeling: A Time Series Analysis of Nepal Stock Exchange

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Keywords: NEPSE Index, Time Series Forecasting, ARIMA Model, Box–Jenkins Methodology, Stationarity

The proposed paper aims to analyze and forecast the NEPSE Index using univariate time-series analysis and the Autoregressive Integrated Moving Average (ARIMA) model. The study uses weekly closing values of the NEPSE Index covering from June 2010 to August 2026, comprising 823 weekly observations. Descriptive statistics were used to summarize the characteristics of the series, while the Augmented Dickey–Fuller test was applied to assess stationarity. In order to identify the model, the autocorrelation and partial autocorrelation were analyzed. Candidate ARIMA models were estimated and compared using information criteria, followed by residual diagnostic testing and evaluation of forecast accuracy.

The raw NEPSE Index series was found to be non-stationary (ADF test, $p = 0.33$), becoming stationary after first differencing. Automated order selection on the raw series suggested ARIMA(0,1,0), but residual diagnostics revealed significant remaining autocorrelation (Ljung–Box, $p = 0.006$), indicating an inadequate fit. Following a log transformation, ARIMA(2,1,0) with drift was identified as the best-fitting model (AIC = -3359.77, BIC = -3340.93), with residual diagnostics confirming no significant autocorrelation remained (Ljung–Box, $p = 0.97$). The model achieved a mean absolute percentage error (MAPE) of 3.06% on out-of-sample data. The twelve-week-ahead forecast indicated a mild upward trend, from approximately 2,649 to 2,706 points, with forecast intervals widening at longer horizons. An alternative ARIMA(1,1,1) with drift specification produced nearly identical forecasts, confirming the stability of the results.

These findings support the use of log-transformed ARIMA models for short-term forecasting of the NEPSE Index and provide a useful reference for investors and policymakers.

T053

Predicting Mental Health Symptoms Using Everyday Digital Behavior: A Statistical Analysis

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Keywords: digital behavior, screen time, social media use, night-time phone use, depression, anxiety, PHQ-9, GAD-7, multiple linear regression, t-test, ANOVA, Chi-square

The growing use of smartphones and social media has drawn increasing attention to the relationship between everyday digital behavior and mental health. This study examined the association between digital behavior and depressive and anxiety symptoms among 392 individuals aged 18 years and above using a quantitative cross-sectional design. Data were collected through an online questionnaire. The Screen Time Index (STI), Social Media Index (SMI), and Night-Time Risk Index (NTI-R) were constructed from various data collected. Data were analyzed using descriptive statistics, correlation analysis, multiple linear regression, *t*-tests, ANOVA, and chi-square tests. Results showed that all three digital-behavior indices were significantly and positively associated with depressive and anxiety symptoms. SMI and NTI-R were significant positive statistical predictors of both outcomes, whereas STI was not an independent predictor. The regression models explained 16.0% of the variance in PHQ-9 scores and 14.5% in GAD-7 scores. Female respondents reported higher depressive and anxiety symptom scores, while relationship status was significantly associated with both outcomes; residence showed no significant association. These findings suggest that social media engagement and night-time phone use may be more strongly associated with mental health symptoms than overall screen time. However, due to the cross-sectional design, the findings examine associations rather than clinical diagnoses.

T054

Empirical Evaluation of Sorting Algorithm Performance Under Heterogeneous Processor Architectures: A Statistical Comparison

Ayush Singh[1], Saksham Pant[1], Arnav Shrestha[1], Laxmi Bhattarai[1]

[1]

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Keywords: paired t-test, Wilcoxon signed-rank test, effect size, Cohen's d, p-value, null hypothesis, bubble sort, merge sort, quicksort

This study compares the runtime efficiency of three sorting algorithms - bubble sort, merge sort, and quicksort - across two different processors (Apple M3, Intel Core Ultra 7). Each algorithm was tested on identical arrays of different sizes arranged in three input orders (ascending, descending, and random), yielding paired timing measurements for direct comparison between processors. A paired t-test was used as the primary statistical method, since each pair of measurements originated from the same array run on both processors rather than from independent samples. To validate these results while accounting for outliers, the Wilcoxon signed-rank test was also applied; agreement between the two tests strengthens confidence in the findings. Finally, effect size (Cohen's d) was calculated to determine the magnitude of the observed performance differences and whether they are practically meaningful.

T055

Comparative Spectral Analysis of Network Robustness Under Node and Edge Failures

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1

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Keywords: spectral graph theory, network robustness, algebraic connectivity, spectral radius, natural connectivity, Laplacian and Adjacency matrix

This study investigates the use of spectral graph properties for characterizing the robustness of complex networks under structural failures. Networks with different topological characteristics will be subjected to random and targeted node and edge removals at varying levels of failure intensity. Algebraic connectivity, spectral radius, and natural connectivity will be evaluated using the eigenvalues of the graph Laplacian and adjacency matrices. Their changes under different failure scenarios will be compared with conventional network robustness measures, including the size of the largest connected component, average path length, and global efficiency. Statistical analysis will be employed to examine the relationship between spectral degradation and structural network damage across different network topologies and failure mechanisms. The study aims to identify spectral measures that provide consistent characterization of network degradation and to assess their effectiveness under different types and intensities of failures. The findings are expected to provide a computational framework for evaluating network robustness through spectral properties and may provide a foundation for identifying structurally critical nodes and edges.

T056

Can Data Provenance and Source Reliability Improve the Reliability of Generative AI?

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Keywords: Generative AI, Large Language Models, Source Reliability

Generative AI systems increasingly use information from a wide range of sources, including academic publications, government websites, news articles, online content, and AI-generated material. While having access to more information can improve the usefulness of these systems, sources can differ greatly in their accuracy, reliability, and independence. When unreliable information is accepted as trustworthy, AI-generated responses can spread misinformation and lead users to make poorly informed decisions. As AI-generated and duplicated content becomes more common, it is becoming increasingly important for AI systems to distinguish reliable evidence from misleading or repeated information. While existing research has extensively examined AI reliability, misinformation, synthetic data, and model degradation, less is known about how generative AI systems weigh conflicting evidence when information about its sources is available. In particular, it remains unclear whether models can prioritize a smaller amount of reliable and independent evidence over a larger amount of unreliable or repeated information. The study aims to explore whether providing AI with information about the origin and reliability of its sources can improve the accuracy of its responses. The study will use two main methods: a literature review and a controlled experiment. The literature review will examine existing research on generative AI reliability, misinformation, source credibility, data provenance, synthetic data, and model collapse. The experiment will use commercially available generative AI models, including DeepSeek, GPT, and Gemini, through a context-injection approach. The models will be given the same factual questions and sets of supporting information, while the information provided about those sources will be changed across three experimental conditions: (1) sources presented without additional information, (2) sources labelled by type and origin, (3) and sources provided with information about their reliability and independence. The questions will focus on general factual topics with answers that can be independently verified. AI responses will be evaluated using factual accuracy, error rate, and self-reported confidence on a defined scale. Confidence will be treated as the model's stated level of certainty rather than access to its underlying probability scores. The study will also examine situations where several sources support the same claim but are copies or derived from the same original source, testing whether AI systems distinguish between the number of sources and the amount of independent evidence. Based on the findings, the study will also consider a potential direction for developing more reliable and trustworthy AI systems, in which models are designed to evaluate source type, provenance, reliability, and independence before relying on information. Such a source-aware approach could provide a basis for future retrieval or training methods that encourage AI systems to prioritize the quality and independence of evidence rather than simply the quantity of available information.

T057

Comparative Performance of Value-at-Risk and Expected Shortfall Under Heavy-Tailed and Volatile Market Conditions: A Statistical Backtesting Approach

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Keywords:

This study aims to compare the performance of Value-at-Risk (VaR) and Expected Shortfall (ES) in measuring market risk in the Nepalese stock market under heavy-tailed and volatile market conditions. Daily price data from the NEPSE Index will be used to calculate returns and examine their statistical characteristics, including volatility, skewness, and kurtosis. Historical, Normal distribution, and Student's t-distribution based methods will be used to estimate VaR and ES at 95% and 99% confidence levels. The models will then be evaluated using rolling-window backtesting by comparing predicted losses with actual market losses. The study will also examine whether model performance changes during periods of high volatility and extreme market movements. The findings are expected to show which risk measurement approach provides more reliable estimates of extreme losses and whether Expected Shortfall provides additional information about tail risk compared with VaR in the Nepalese stock market.

T058

Structural Uncertainty in Constructed Graphs: A Message-Passing Scheme for Incomplete Data

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Keywords: Graph neural networks, Structural uncertainty, Graph construction, Missing data, Neutrosophic representation

Graph neural networks (GNNs) are increasingly employed for incomplete tabular data without an inherent graph, requiring a graph to be constructed from feature similarity and used as a fixed input. In this regard, neutrosophic graph reconstruction is utilized to represent the uncertainty introduced during construction. Different similarity definitions may select different edges from the same data, a rule may change its edge selections after small feature perturbations, and some record pairs may have too few originally observed features for reliable comparison. Existing pipelines often ignore these sources or aggregate them into a single indeterminacy value. Using four density-matched construction rules and repeated feature subsampling, each candidate edge is described by three measured quantities: support across rules; a signed measure of whether disagreement between rules or instability within a rule dominates; and an observability measure of the information available before imputation. These quantities are measured from the data rather than assigned manually. During message passing, support determines how much information is transmitted through an edge, while the source of uncertainty determines how that information is transformed. The neutrosophic mapping evaluated here combines disagreement, instability, and limited observability into a single indeterminacy value, whereas the source-conditioned form preserves their distinctions. Four groups will be compared under increasing

levels of missing data: non-graph methods, conventional GNNs on constructed graphs, an aggregated- neutrosophic GNN, and the source-conditioned scheme examined here. These comparisons will assess

whether preserving the source of structural uncertainty, rather than collapsing it into a single value, can improve classification reliability as data become increasingly incomplete.

T059

AI-Powered Acoustic Waste Classification for Automated Waste Segregation Using a Lightweight 1D-CNN

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Keywords: Acoustic waste classification, Automated waste segregation, 1D-CNN, Mel-Filterbank Energy, Edge AI, ESP32, Smart waste management

Graph neural networks (GNNs) are increasingly employed for incomplete tabular data without an inherent graph, requiring a graph to be constructed from feature similarity and used as a fixed input. In this regard, neutrosophic graph reconstruction is utilized to represent the uncertainty introduced during construction. Different similarity definitions may select different edges from the same data, a rule may change its edge selections after small feature perturbations, and some record pairs may have too few originally observed features for reliable comparison. Existing pipelines often ignore these sources or aggregate them into a single indeterminacy value. Using four density-matched construction rules and repeated feature subsampling, each candidate edge is described by three measured quantities: support across rules; a signed measure of whether disagreement between rules or instability within a rule dominates; and an observability measure of the information available before imputation. These quantities are measured from the data rather than assigned manually. During message passing, support determines how much information is transmitted through an edge, while the source of uncertainty determines how that information is transformed. The neutrosophic mapping evaluated here combines disagreement, instability, and limited observability into a single indeterminacy value, whereas the source-conditioned form preserves their distinctions. Four groups will be compared under increasing levels of missing data: non-graph methods, conventional GNNs on constructed graphs, an aggregated-neutrosophic GNN, and the source-conditioned scheme examined here. These comparisons will assess whether preserving the source of structural uncertainty, rather than collapsing it into a single value, can improve classification reliability as data become increasingly incomplete.

T060

Assessing Human Development through Life Satisfaction: Evidence from Bagmati Province of Nepal

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Keywords: Human development, Life satisfaction, Subjective well-being, Livelihood, Quality of life, Technology adaptation

Life satisfaction represents a subjective assessment of overall quality and conditions of life, while human development reflects the expansion of people's capabilities, opportunities, choices, and freedoms to lead healthy, educated, productive, and meaningful lives. This study examines the relationship between life satisfaction and the level of human development among households in the study areas. Data were collected from 350 respondents and assessed based on perceptions of living conditions, health, education, livelihood opportunities, social relationships, access to services, freedom, decision-making, and overall well-being. Human development was assessed through income, education, social governance, and technological adaptation. Approximately 61% of respondents reported high or moderately high life satisfaction, while 54% were classified as having moderate to high human development. Respondents with better access to healthcare, education, livelihood opportunities, income, technology, and decision-making opportunities generally reported higher life satisfaction. The findings indicate a positive relationship between human development and life satisfaction, suggesting that improved capabilities, living conditions, and access to essential services are generally associated with greater life satisfaction. However, life satisfaction may vary according to individual expectations, social relationships, aspirations, and personal circumstances. The study concludes that human development should be assessed not only through objective socioeconomic conditions but also through people's subjective experiences of well-being. Life satisfaction provides an important complementary dimension for understanding the outcomes of human development. Integrating human development indicators with life satisfaction can provide a broader understanding of people's living conditions and well-being and support policies aimed at improving livelihoods, social inclusion, quality services, governance, technology access, and meaningful opportunities for a better quality of life in Nepal.

3 Conference Programme

Day 1 – 19 September 2026

Time	Activity
08:30–09:30	Registration
09:30–10:00	Inaugural Session at **CV Raman Hall**
10:00–11:00	**Keynote Lecture I** **Title:** Phase-wise Analysis of the Bowling Performance of Cricketers in the Nepal Premier League **Speaker:** Dibyojyoti Bhattacharjee
11:00–11:10	Tea Break
11:10–11:30	**Keynote Lecture II** **Title:** Title **Speaker:** Shankar Prasad Khanal
11:30–13:00	**Parallel Oral Sessions**

Area	Room	Papers
Statistics - Modern Data Science & AI	CV Raman Hall	T001–T006
Statistical Learning to Machine Learning & Big Data	Senate Hall	T007–T012

Time	Activity
13:00–14:00	Lunch Break
14:00–15:30	**Parallel Oral Sessions**

Area	Room	Papers
Environment, Climate Change & Sustainability	CV Raman Hall	T025–T030
Emerging Trends in Bayesian Methods	Senate Hall	T031–T036

Time	Activity
15:30–16:00	Tea Break
16:00–17:30	Poster Presentation Session

Day 2 – 20 September 2026

Time	Activity		
09:00–10:00	**Keynote Lecture II** at **CV Raman Hall**		
	Title: Title		
	Speaker: Sudan Jha		
10:00–10:20	Tea Break		
10:20–10:40	**Invited Lecture II**		
	Title: Title Not Found		
	Speaker: Author Not Found		
10:40–12:10	**Parallel Oral Sessions**		
	Area	**Room**	**Papers**
	New Methodologies and Recent Progress	CV Raman Hall	T049–T054
	Statistics in Risk Management	Senate Hall	T055–T060
12:10–13:10	Lunch Break		
13:10–14:40	**Parallel Oral Sessions**		
	Area	**Room**	**Papers**
	Public Health and Bio Statistics	CV Raman Hall	T067–T072
	Statistics in Management Sciences	Senate Hall	T073–T078
14:40–15:00	Tea Break		
15:00–16:00	Valedictory Session and Closing Ceremony at **CV Raman Hall**		

4 Technical Papers

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